

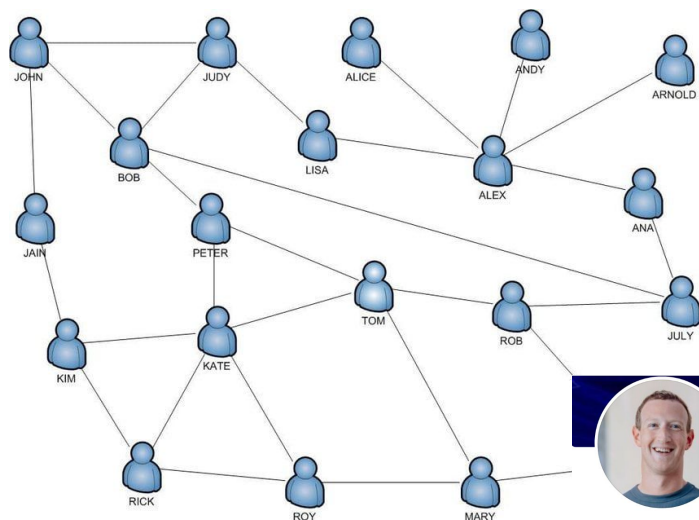
GeoMedia: A Social Media for Location-Based and Time-Sensitive Content Sharing

Alberto Morini
(2107783)

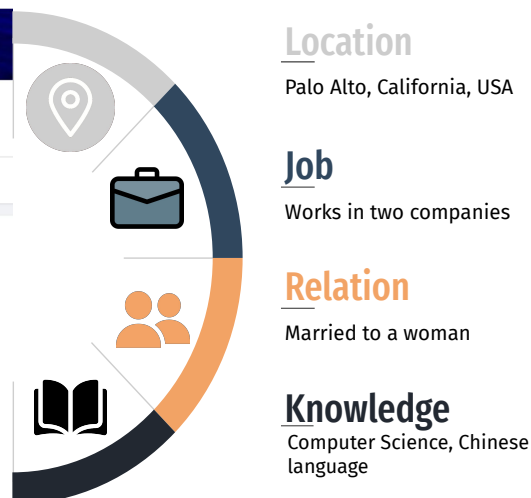
Supervisor: Prof. Claudio Enrico Palazzi
Co-supervisor: Dott. Lorenzo Perinello

Padua, 24th September 2026

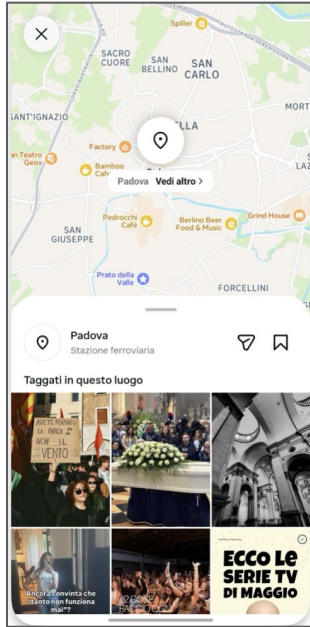
Online Social Networks



- OSNs profile-centric
- Real life representation
- Profile and content suggestions



Location based apps

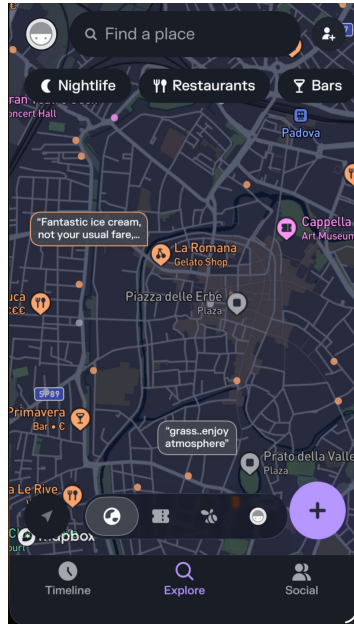


Instagram

Location constraint

Creation: Free

Access: Free

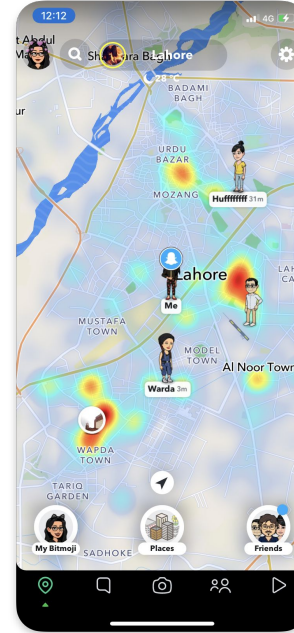


Swarm (Foursquare)

Location constraint

Creation: Free

Access: Free

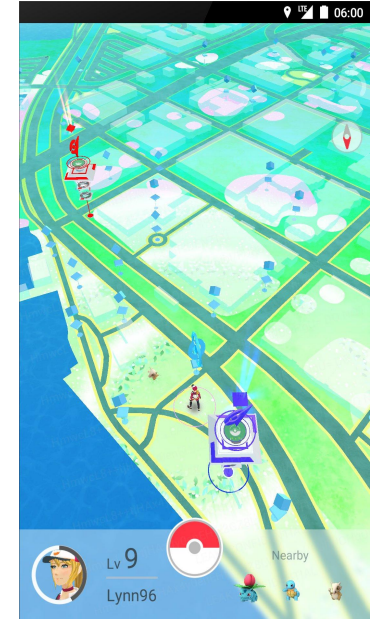


Snapchat

Location constraint

Creation: Current

Access: Free



Pokémon GO

Location constraint

Creation: N.A.

Access: Current

An open-source social network framework centered on content, bridging the physical and digital worlds by anchoring multimedia posts to real-world geographic coordinates, where they can be accessed.

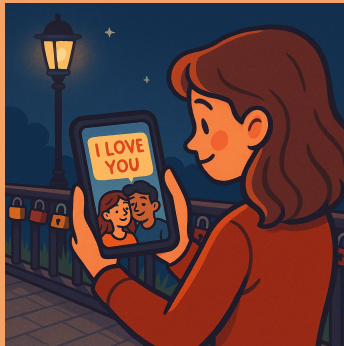
Geographical limited



Interactive experiences

- *Museum visits*
- *Educational tours*

Time sensitive



Personal messages

- *Recurrency visibility*
- *Account-based access*

Sequentiality



Location based games

- *Scavengers hunts*
- *Extended escape room*

Example



Science4All



Math
problems

Viewers: all
Area: 50 meters



Cybersec
news

Viewers: all
Area: 15 meters
Time: 26 sept - 26 sept

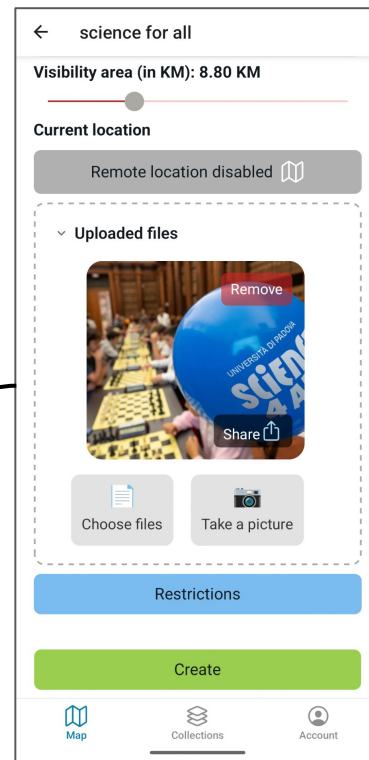


AI chess

Viewers: Elo level +1400
Area: 100 meters

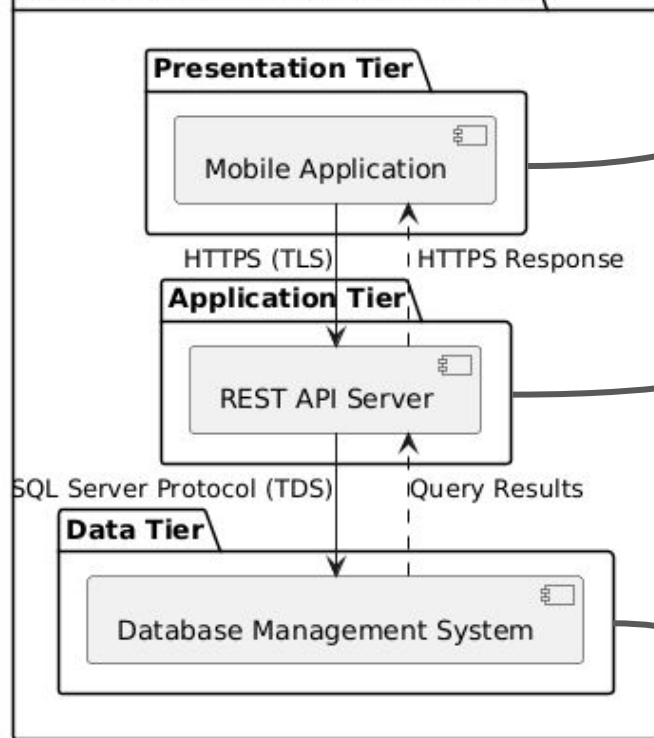
Time limit: 25 sept - 27 sept

Creators: Teachers, students



Software architecture

GeoMedia - Software Architecture



React Native

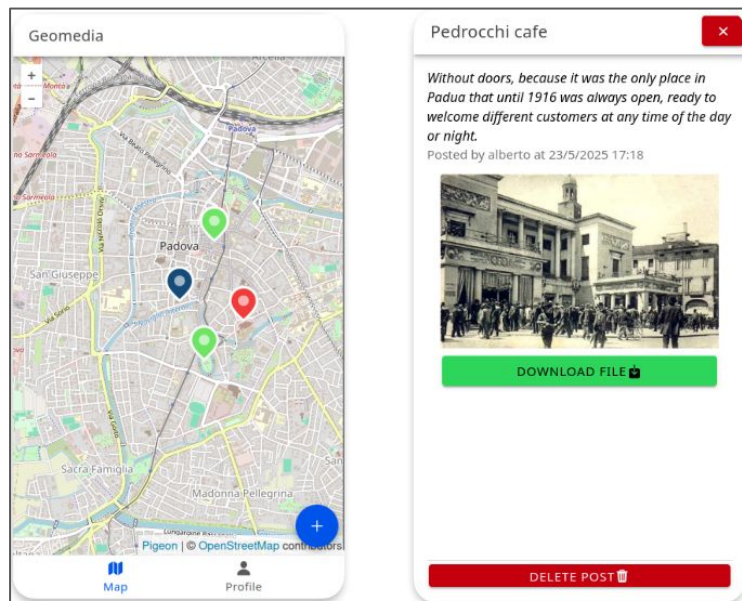


Python with FastAPI

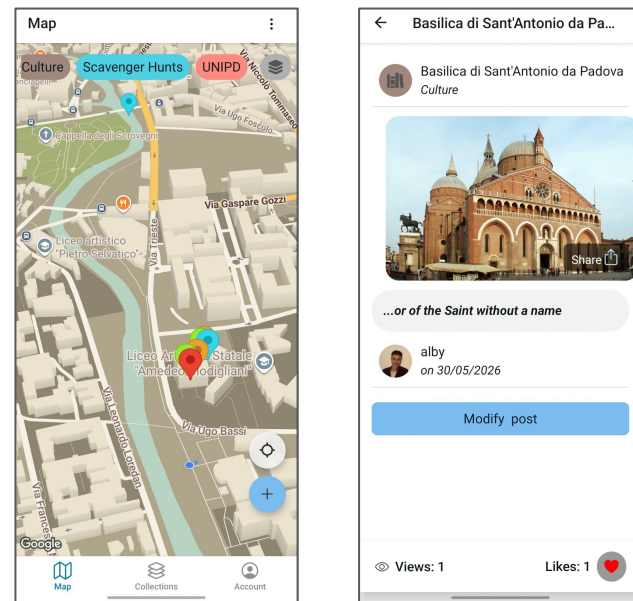


Microsoft SQL Server

Mobile app frameworks



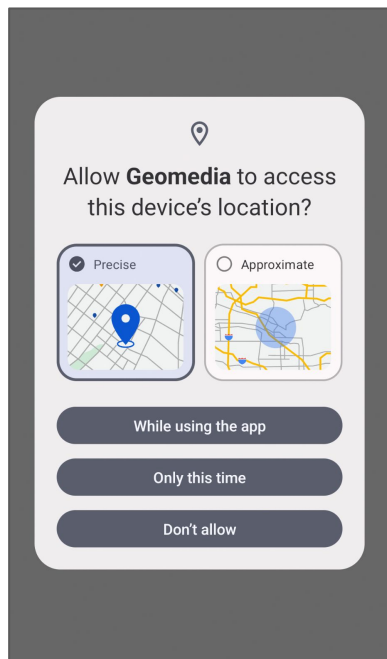
Proof of concept



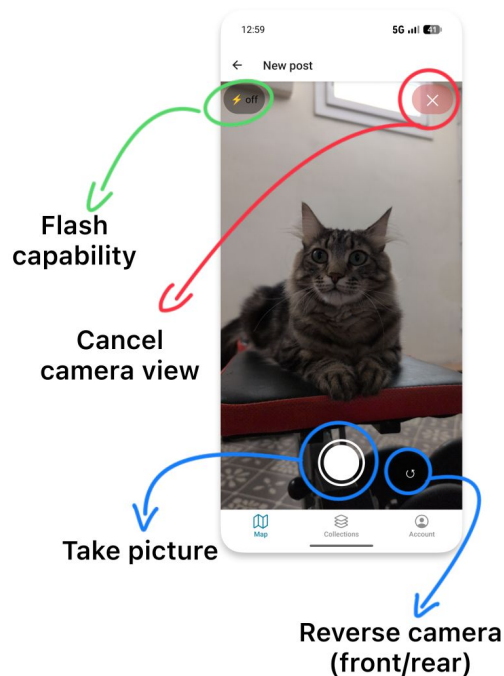
Official application

vs.

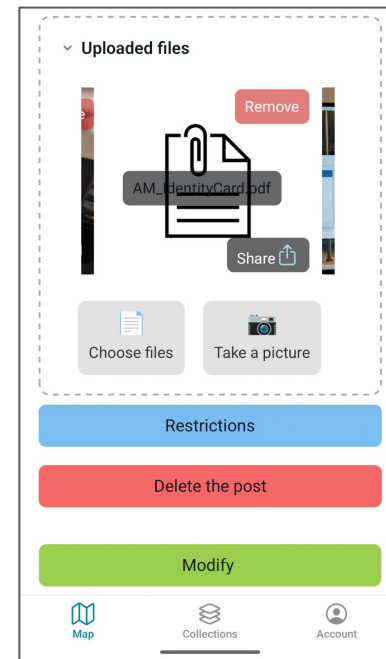
Device integration



Device location

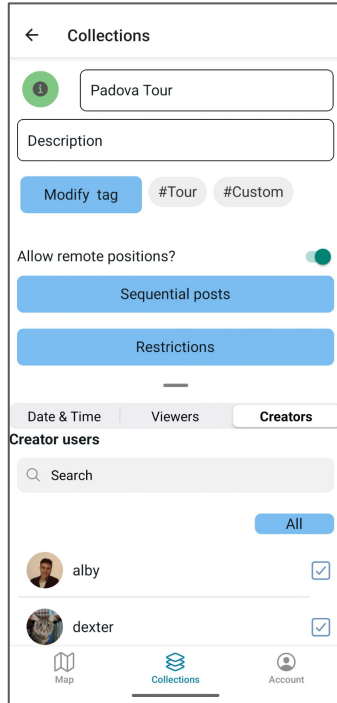


Built-in camera

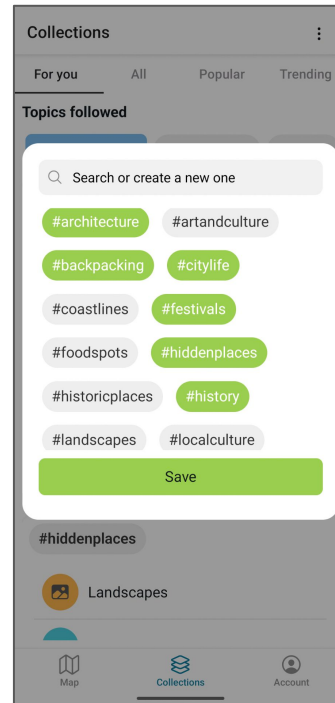


File uploading

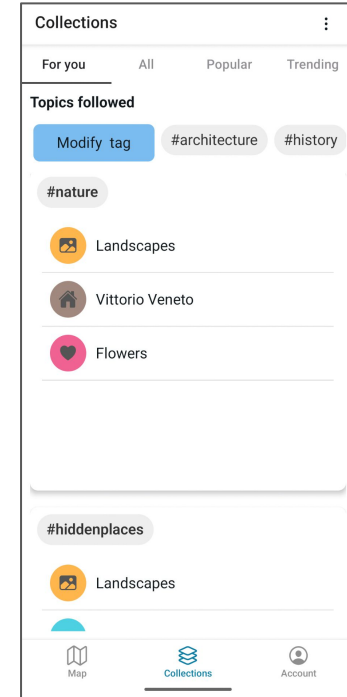
Collections



Creation & limitations

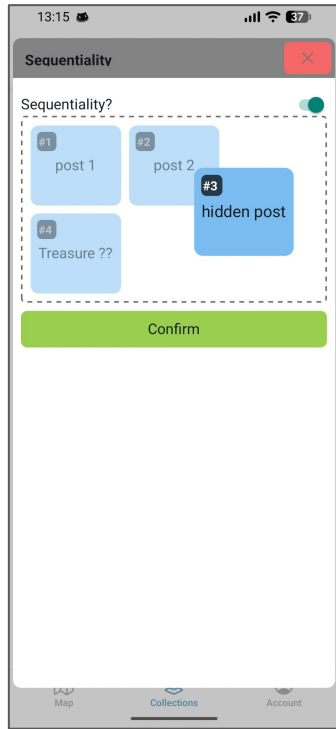


Topics



For you section

Sequentiality



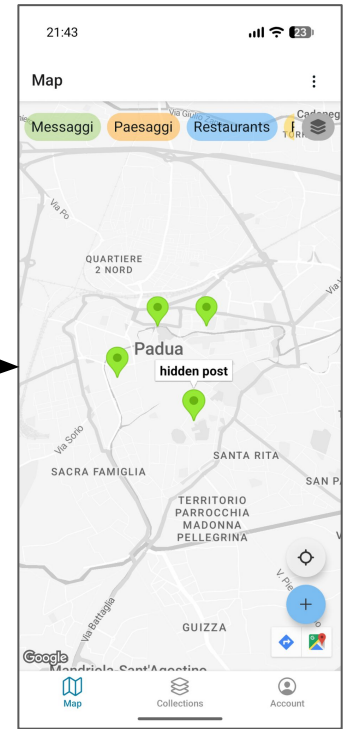
Definition



Start

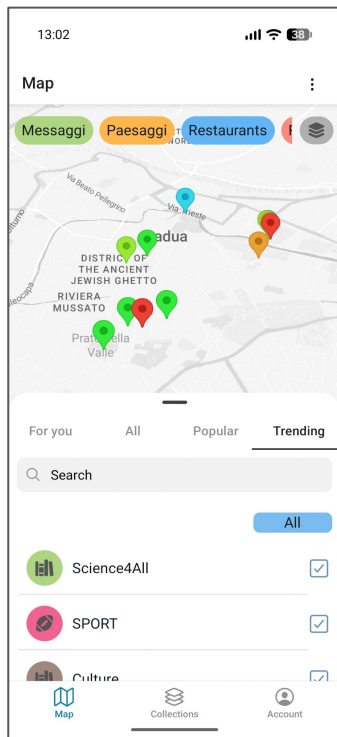


#3 unlocked

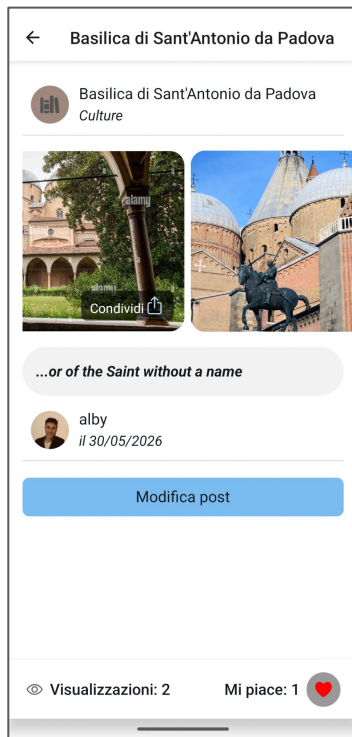


#4 unlocked

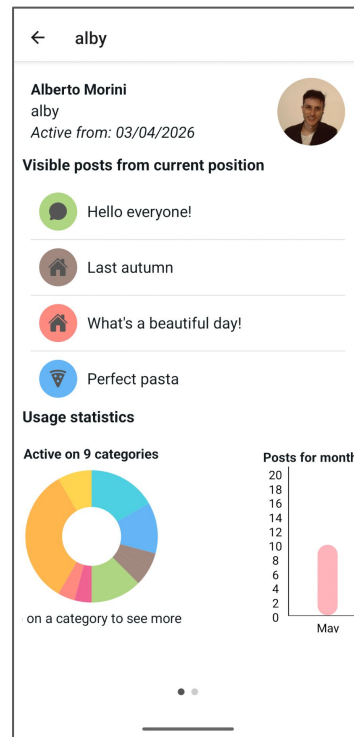
Results



Map exploration



Post viewer



Profile viewer

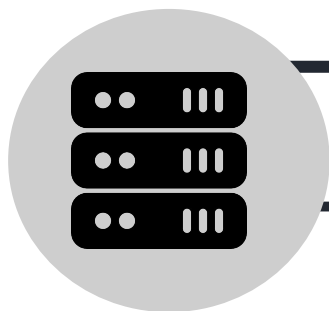
```
1 app = FastAPI(title="Geomedia API",dependencies=[Depends(verify_auth)])
2
3 if __name__ == "__main__":
4     print(f"Server started on port: {PORT}")
5     uvicorn.run(app, host="0.0.0.0", port=PORT,
6                 ssl_keyfile="/etc/letsencrypt/live/geomediasrv.duckdns.org/privkey.pem",
7                 ssl_certfile="/etc/letsencrypt/live/geomediasrv.duckdns.org/fullchain.pem",
8                 )
```

RESTful API

- GET
- POST
- DELETE

protocol://domain.name:port/path?query

https://geomediasrv.duckdns.org:9911/post/11



File systems

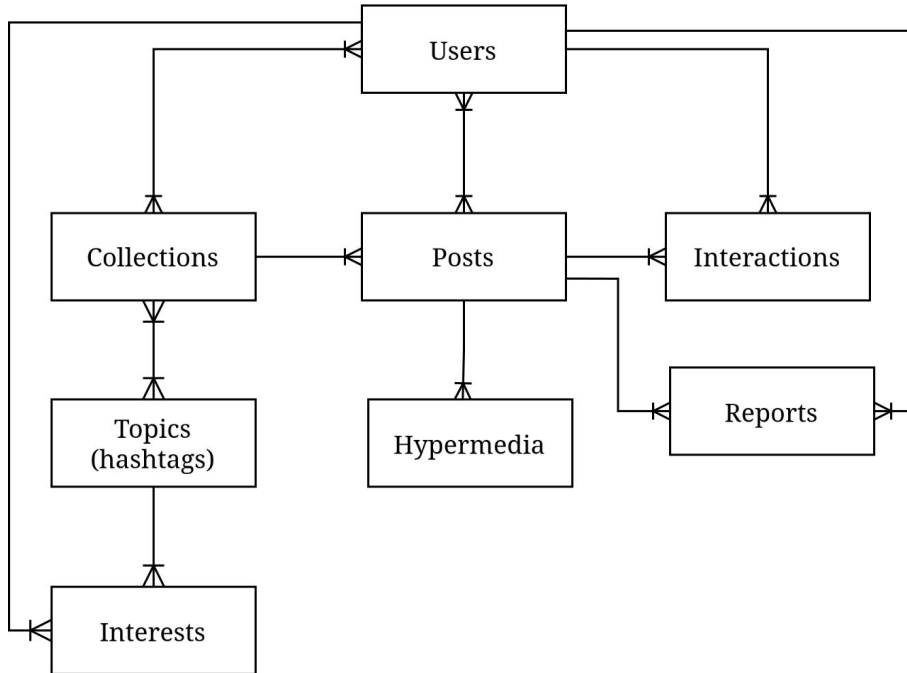


Databases



Docs

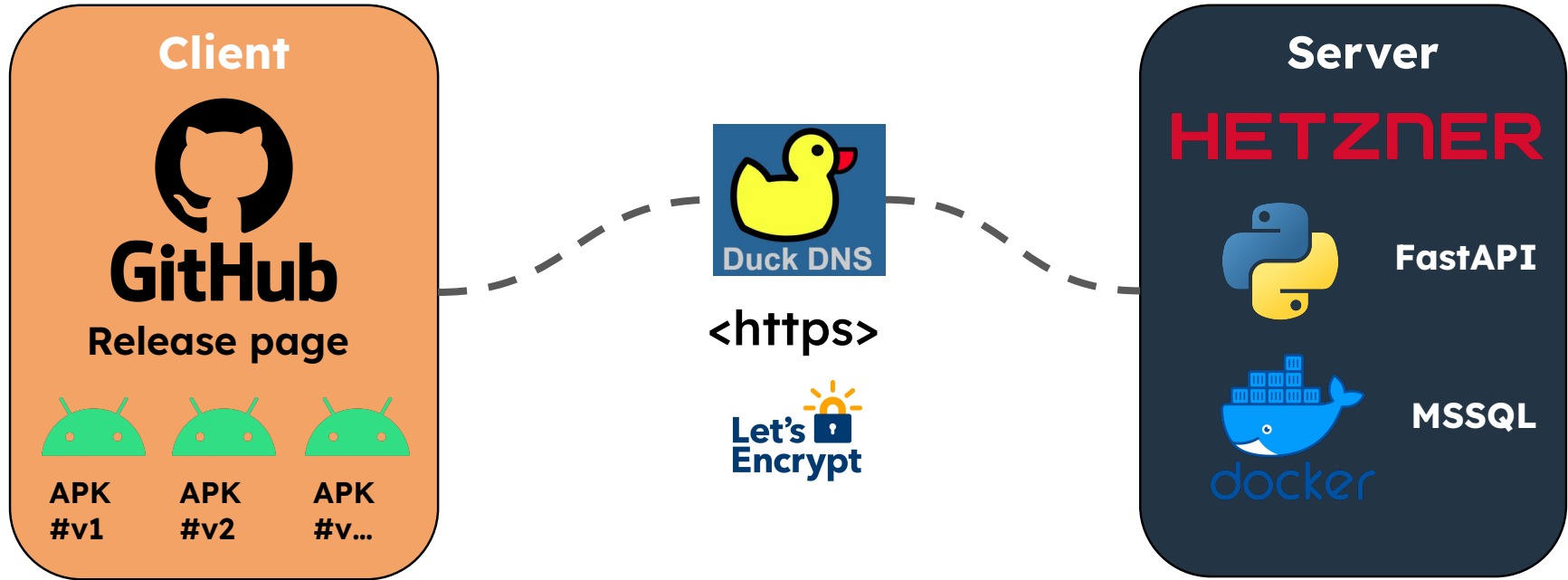
ER schema



Fundamental principles

- **Normalization:** reduce redundancy, avoid ambiguities
- **Referential integrity:** primary and foreign key
- **ACID properties:** ensure reliable data transactions
- **Data integrity:** constraint

Deployment



Test results

15 candidates

Education

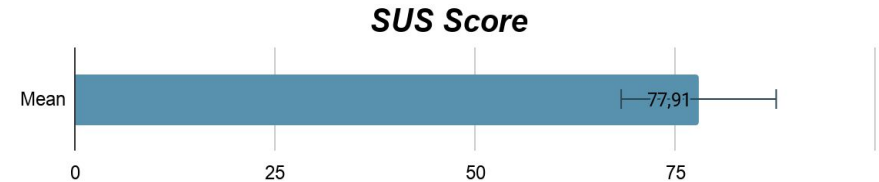
- 20% | no IT background
- 33% | CS students
- 47% | IT professionals

Age

- 33% | 20-29
- 13% | 30-39
- 13% | 40-49
- 33% | 50-59
- 7% | 60+

System Usability Scale questionnaire

- **Mean** = 77.91
- **SD** = 12.34

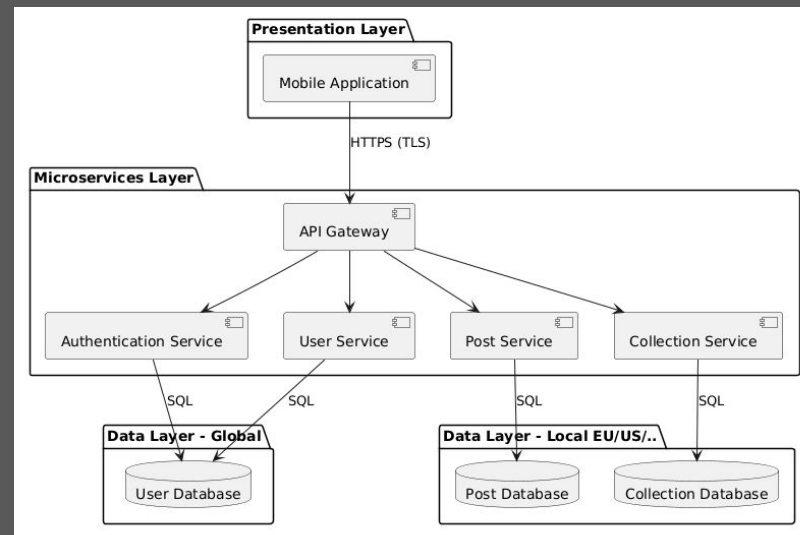


Future developments

Other features

- Time capsule with NFC
- Altitude integration
- Augmented Reality
- Mesh networks
- QR codes adoption
- Implementation of comments
- Post suggestions

Microservice adoptance



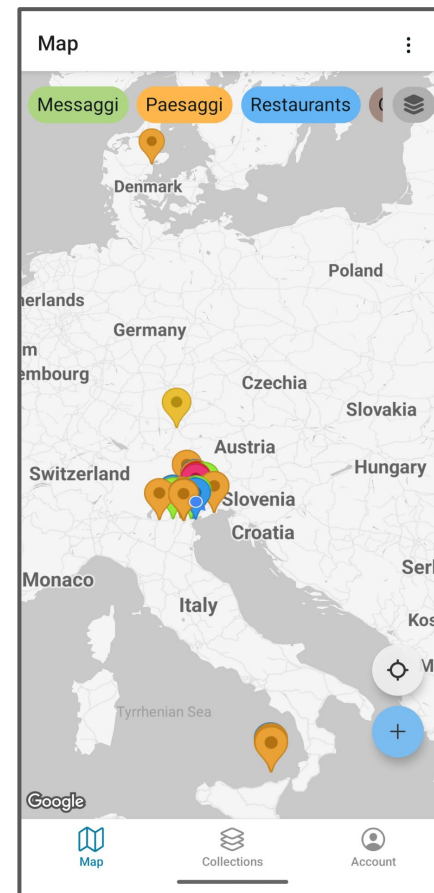
Thanks.

Alberto Morini
(2107783)



Why GeoMedia?

- Content centric application
- Real-world integration
- Different use cases
- Mobile first approach
- Accessible UX
- Open source project
- Privacy oriented
- Cross platform



Unique features

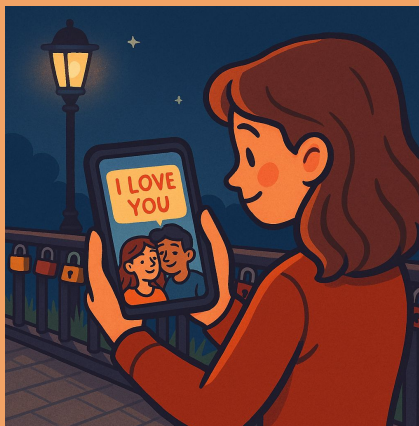
Location-restriction



Interactive experiences

- *Local guides*
- *Educational tours*

Time sensitive



Personal messages

- *Recurring visibility*
- *Visibility and creation scope*

Sequentiality



Location based games

- *Scavenger hunts*
- *Real-time adventure*

Item	Statement	Mean	SD
1	I think that I would like to use this system frequently.	2.54	1.21
2	I found the system unnecessarily complex.	1.54	0.78
3	I thought the system was easy to use.	3.61	1.21
4	I think that I would need assistance to use the system.	1.60	0.70
5	I found the various functions in the system were well integrated.	4.00	0.82
6	I thought there was too much inconsistency in the system.	1.70	0.82
7	I would imagine that most people would learn to use the system very quickly.	4.10	0.74
8	I found the system very cumbersome to use.	1.50	0.71
9	I felt very confident using the system.	4.20	0.79
10	I needed to learn a lot of things before I could get going with the system.	1.80	0.92

Table 8.2: Analysis of SUS questionnaire responses.

<i>Mechanism</i>	Description
Account verification (OTP)	During the registration phase, the system verifies that the provided email address belongs to the user registering the account.
HTTPS/TLS	It protects data exchanged between clients and backend services, applying encryption.
HMAC	Used to verify the authenticity and integrity of messages and requests and combined with nonces or timestamps, to mitigate replay and masquerade attacks.
HASH (MD5)	Used to hash passwords into a fixed-length representation before storage.
Prepared Statement	Implemented to mitigate SQL injection attacks by preventing user input from being interpreted as part of SQL queries.
Content moderation	Manual but designed to support automated moderation using LLMs and human review to identify and manage potentially harmful content.

Table 6.1: Cybersecurity mechanism implemented.

GET	/profile/pfp	Profile Getpfp	▼
GET	/profile/info/{profile_id}	Profile Getinfo	▼
GET	/profile/stats	Profile Get Stats	▼
POST	/profile	Profile Editinfo	▼
GET	/profile/show_allowed_post	Profile Show Allowed Post	▼
GET	/users	Users List	▼
POST	/profile/interests	Profile Interest Merge	▼
GET	/profile/interests	Profile Interest Get	▼

COLLECTION



POST	/collection	Collection Merge	▼
GET	/collection/hashtags	Hashtag Get	▼
GET	/collection/id/{collection_id}	Collections Get Fullcollection	▼
GET	/collections	Collections Get Fullcollection	▼
GET	/collection/posts	Collection Posts Get	▼
GET	/collection/groupByTag	Collections By Tag	▼
GET	/collections/get_local	Collections Get Local	▼

POST



POST	/post	Post Merge	▼
GET	/post/id/{post_id}	Get Post	▼

<i>Feature</i>	GeoMedia	Jagat	Aulinn	Swarm	Snapchat	Instagram	Google Maps
Arbitrary location posting	✓	✓	✓	–	–	–	–
Posting from remote	Collection dependent	✓	✓	✓	–	✓	✓
Location-based restrictions	✓	–	–	–	–	–	–
Time restrictions	✓	–	–	–	✓	✓	–
Recurrent post	✓	–	–	–	–	–	–
Sequential posts	✓	–	–	–	–	–	–
Multi-author collections	Selective contributors	–	–	–	–	Everyone	Everyone
User-based restrictions	Selective viewers	–	–	–	Selective viewers	Selective viewers	–
Topic-based content	✓	–	–	✓	–	–	✓
User statistics	✓	–	–	✓	–	–	✓

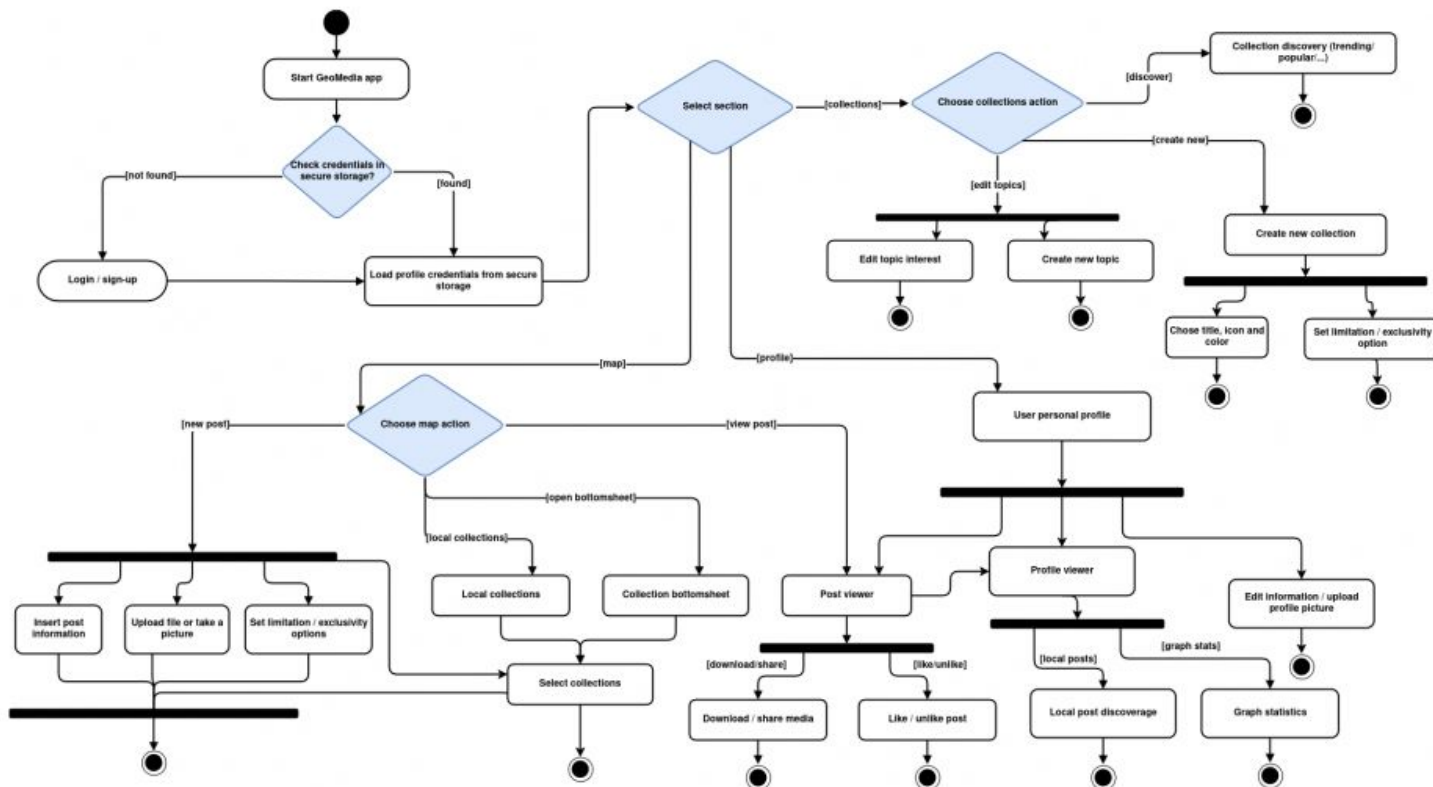


Figure 5.4: Activity diagram of GeoMedia app.

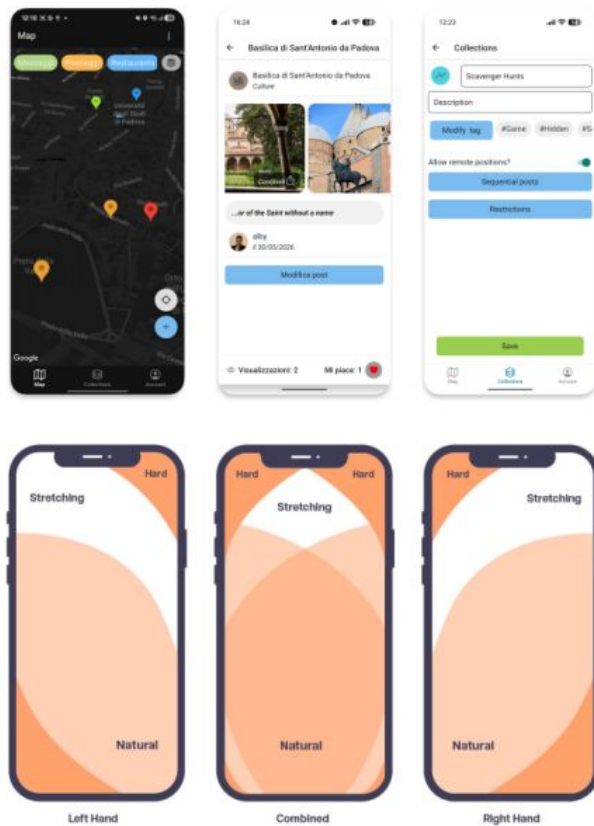


Figure 5.5: GeoMedia thumb-rule comparison [81].

Releases

Tags

Release list

GeoMedia 1.5.4

GeoMedia 1.5.3

GeoMedia 1.5.2

GeoMedia 1.5.1

GeoMedia 1.5

GeoMedia 1.4.2

GeoMedia 1.4.1

GeoMedia 1.4.0

GeoMedia 1.3

GeoMedia 1.2

< Previous

Next >

GeoMedia 1.5.4

Compare

albertomorini released this jul 24

1.5.4

4477f1c

- Fixed post view for navigation bar with button.
- If no collection are settings, the locale one are automatically activated

Full Changelog:

1.5.3...1.5.4

Assets

GeoMedia 1.5.3

Compare

albertomorini released this jul 23

1.5.3

4c56dc3

Full Changelog:

1.5.2...1.5.3

- Improvement and minor bug fix

Assets

Figure 8.1: GitHub release page of GeoMedia app.

17:20

4B

Settings

Get in touch

Email Me

Telegram

Github

Language

English

Map style

System default

Config backend server

Protocol

https

Server name (or IP)

geomediasrv

Port

9911

Confirm

Logout

Logout

Generate Signed App Bundle or APK

Module: Geomedia.app

Key store path: /home/alby/Android/morinialberto

Key store password:

Key alias: key0

Key password:

☒ Remember passwords

Previous Next Cancel

New Key Store

Key store path: /home/alby/Android/morinialberto

Password: Confirm:

Key

Alias: key0

Password: Confirm:

Validity (years): 25

Certificate

First and Last Name: Alberto Morini

Organizational Unit: ComputerScience

Organization: University of Padua

City or Locality: Padua

State or Province: Veneto

Country Code (XX): IT

OK Cancel

12:18

← Collections



Scavenger Hunts

Description

Modify tag #Game #Hidden #S

Allow remote positions? ☒

Sequential posts

Confirm

Date & Time Viewers Creators

The post will be visible from:

7/26/2026 12:18:00 PM

The post will not be visible after:

7/31/2026 12:18:00 PM

Recurrent?

Never Monthly Yearly

Map

Collections

Account

17:20

← Settings

Get in touch

Email Me Telegram Github

Language

English

Map style

System default

Config backend server

Protocol

https

Server name (or IP)

geomediasrv

Port

9911

Confirm

Logout

Logout

11:39

← Prato della valle



Prato della valle
Paesaggi

Report

Choose the kind of report

Report reason (at least 100 characters)

Confirm

 on 30/05/2026

Report post

Views: 3 Likes: 1

11:37

← Prato della valle



Prato della valle
Paesaggi

Choose the kind of report

Wrong places/information

Illegal content

Hate speech

Harassment / Bullying

Adult content

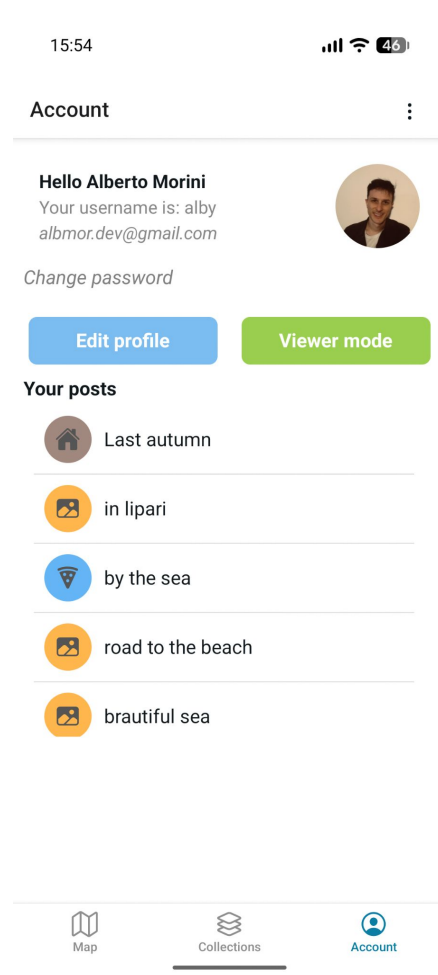
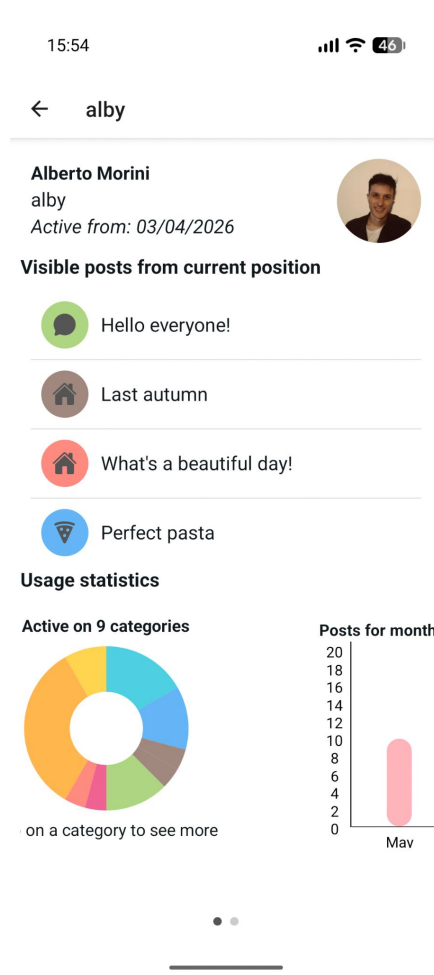
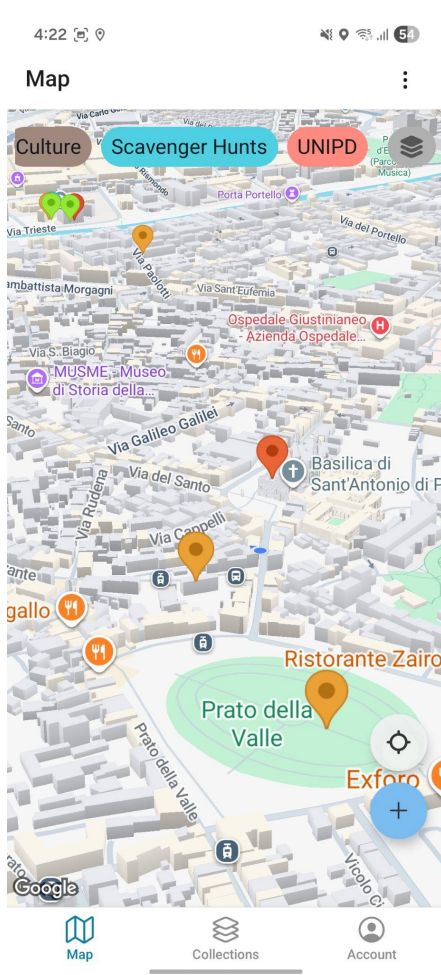
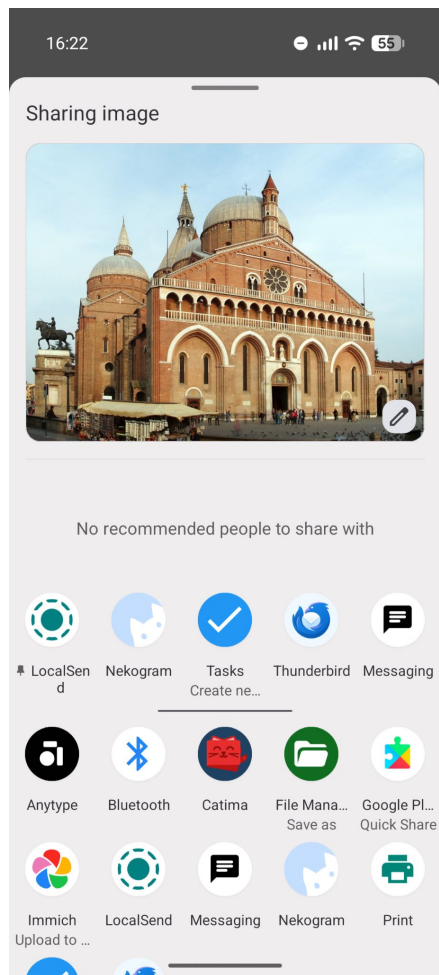
Impersonation

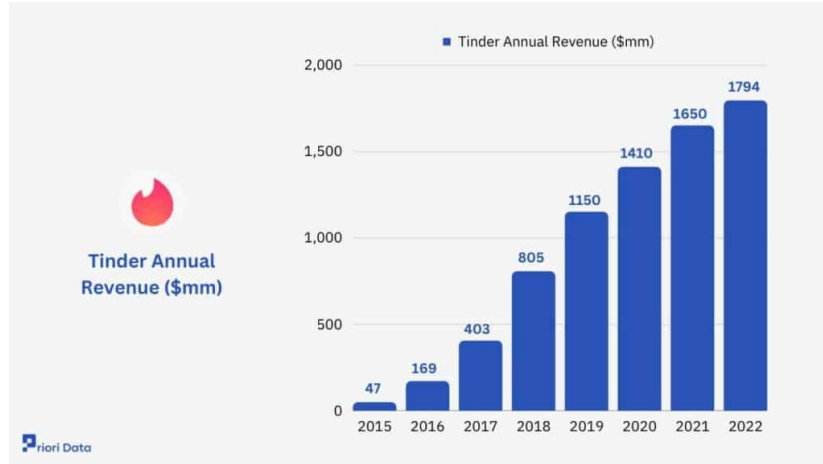
Copyright violation

Privacy violation

Other

Views: 3 Likes: 1





Meta's revenue 2024: \$164.501 billion
 Meta's revenue 2024: \$200.966 billion

